

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I 192325076 – SHAM S. certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	

Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Scenario-Based Research Report

1. Title of AI Application

AI-Assisted Medical Diagnosis Using Large Language Models and Multimodal Artificial Intelligence

Domain: Healthcare

2. Domain Selection

Selected Domain: Healthcare

Artificial Intelligence is revolutionizing healthcare by assisting doctors in disease diagnosis, medical image interpretation, patient monitoring, treatment planning, and clinical decision-making. Among recent innovations, AI-assisted medical diagnosis powered by Large Language Models (LLMs) and multimodal AI has become one of the most significant advancements in modern healthcare.

3. Introduction and Background

Healthcare systems worldwide face numerous challenges, including increasing patient populations, shortages of healthcare professionals, rising medical costs, and diagnostic errors. Traditional diagnostic methods rely heavily on physician expertise and manual interpretation of patient records, laboratory reports, and medical images.

Recent advances in Artificial Intelligence have enabled the development of AI-assisted medical diagnosis systems capable of analyzing medical images, patient symptoms, electronic health records (EHRs), laboratory reports, and clinical notes. These systems provide healthcare professionals with intelligent recommendations, helping improve diagnostic accuracy and reduce the time required for clinical decision-making.

Modern AI applications combine Computer Vision, Natural Language Processing (NLP), Deep Learning, and Generative AI to deliver comprehensive healthcare assistance.

4. Problem Identification

Modern hospitals encounter several problems:

- Increasing number of patients
- Limited availability of specialist doctors
- Delayed diagnosis
- Human errors in medical image interpretation
- Inconsistent treatment recommendations
- High healthcare costs
- Large volumes of Electronic Health Records (EHRs)
- Difficulty in integrating multiple medical data sources

These challenges can delay treatment and negatively affect patient outcomes.

5. Need for AI Solution

Artificial Intelligence addresses these issues by:

- Assisting doctors in disease diagnosis
- Detecting diseases at early stages
- Reducing diagnostic errors

- Providing faster clinical recommendations
- Supporting personalized medicine
- Continuously learning from medical datasets
- Improving healthcare accessibility
- Enhancing hospital efficiency

Unlike traditional systems, AI can analyze thousands of medical records within seconds while maintaining high accuracy.

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Disease Detected:
Pneumonia

Confidence:
96.8%

Affected Area:
Lower Left Lung

Recommendation:
Consult Pulmonologist
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6. AI Technologies Used

The AI-assisted diagnosis system integrates multiple AI technologies.

Machine Learning

- Disease prediction
- Risk assessment
- Patient outcome prediction

Deep Learning

Deep Neural Networks analyze:

- X-rays
- CT scans
- MRI images
- Ultrasound images

Popular architectures include:

- CNN (Convolutional Neural Networks)
- ResNet
- DenseNet
- EfficientNet

Computer Vision

Used for:

- Tumor detection
- Organ segmentation
- Fracture detection
- Skin disease identification

- Lung disease detection
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Natural Language Processing (NLP)

Processes:

- Doctor notes
- Electronic Health Records
- Clinical reports
- Laboratory reports
- Medical literature

Models include:

- BioBERT
 - ClinicalBERT
 - Med-PaLM
 - GPT-based healthcare assistants
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Generative AI

Used for:

- Medical report summarization
 - Clinical documentation
 - Patient question answering
 - Treatment recommendation support
 - Healthcare chatbots
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Reinforcement Learning

Applied in:

- Personalized treatment planning
 - Drug dosage optimization
 - Adaptive healthcare recommendations
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7. Working Mechanism

The AI-assisted diagnosis system operates through several stages.

Patient Visits Hospital



Patient Data Collection

(Symptoms, Medical History, Vitals)



Medical Tests Performed

(Blood Test / X-ray / CT / MRI / ECG)



Data Uploaded to Hospital Information System (HIS)



AI System Receives the Data



Data Preprocessing

- Remove noise
- Normalize images
- Clean patient records
- Extract important features



AI Analysis

- CNN analyzes X-rays/CT scans
- NLP analyzes doctor's notes
- Machine Learning predicts disease probability
- LLM summarizes findings



AI Generates Results

- Possible diseases
- Confidence score
- Highlight affected region
- Suggested additional tests



Doctor Reviews AI Report



Doctor Makes Final Diagnosis



Treatment Plan



Patient Follow-up

Step 1 – Data Collection

The system collects:

- Patient symptoms
- Medical history
- Laboratory reports
- Blood test results
- Medical images
- Vital signs
- Electronic Health Records

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Step 2 – Data Preprocessing

- Remove missing values
- Normalize data
- Image enhancement
- Noise removal
- Feature extraction
- Text cleaning

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Step 3 – AI Processing

Different AI models process different inputs.

Computer Vision analyzes images.

NLP analyzes clinical documents.

Machine Learning predicts disease probability.

Large Language Models summarize clinical information.

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Step 4 – Clinical Decision Support

The AI system generates:

- Possible diagnosis
- Disease probability
- Risk score
- Suggested investigations
- Treatment recommendations

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Step 5 – Doctor Validation

Healthcare professionals review AI recommendations before making the final medical decision.

8. Data Sources and Processing

The system uses multiple healthcare datasets.

Medical Images

- Chest X-rays
- MRI scans
- CT scans
- Histopathology slides

Clinical Data

- Electronic Health Records
- Laboratory reports
- Vital signs
- Patient demographics

Medical Literature

- Clinical guidelines
- Published research papers
- Medical textbooks

Data Processing

- Image preprocessing
 - Data normalization
 - Tokenization
 - Feature engineering
 - Data augmentation
 - Model training
 - Validation
 - Inference
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9. Real-World Implementation

AI-assisted diagnosis is already implemented by several healthcare organizations.

Google DeepMind

Developed AI systems capable of detecting eye diseases and assisting in radiology diagnosis.

Microsoft

Provides AI-powered healthcare solutions through Azure AI for medical imaging and clinical documentation.

Mayo Clinic

Uses AI for disease prediction, radiology analysis, and personalized treatment support.

Cleveland Clinic

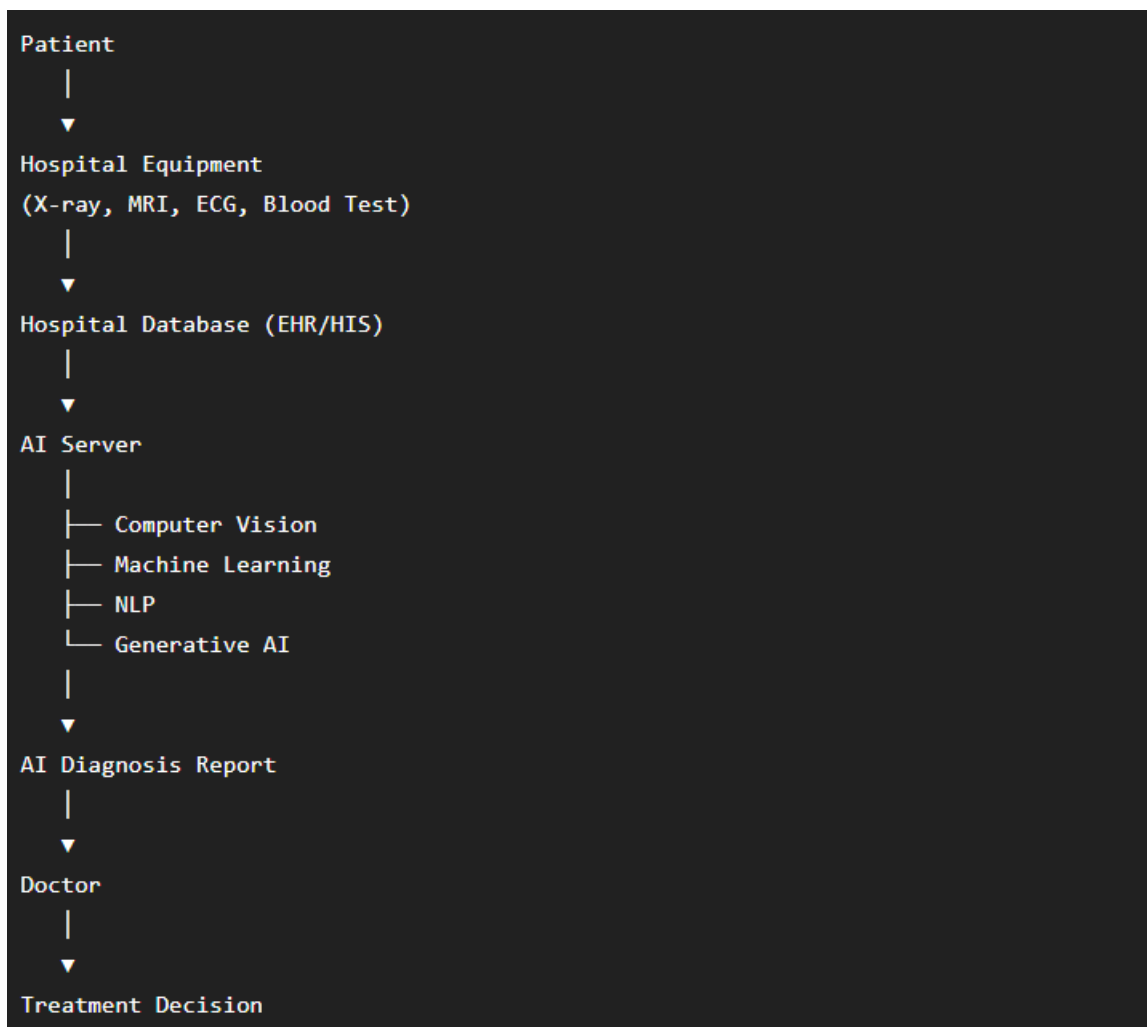
Applies AI for cardiovascular disease prediction and patient care optimization.

Apollo Hospitals (India)

Uses AI-assisted radiology, ICU monitoring, predictive analytics, and clinical decision support.

Tata Memorial Centre (India)

Uses AI in cancer diagnosis and pathology image analysis.



10. Impact and Benefits

AI-assisted medical diagnosis provides significant benefits.

Improved Accuracy

- Early disease detection
- Reduced diagnostic errors

Faster Diagnosis

- Reports generated within seconds
- Reduced patient waiting time

Increased Productivity

- Less administrative workload
- Automated documentation

Personalized Medicine

- Individual treatment recommendations
- Risk prediction

Better Patient Care

- Continuous monitoring
- Improved treatment outcomes

Cost Reduction

- Reduced unnecessary tests
 - Efficient resource utilization
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11. Challenges and Limitations

Despite its benefits, several challenges remain.

Technical Challenges

- Large computational requirements
- Data quality issues
- Model bias
- Limited explainability

Ethical Challenges

- AI fairness
- Bias in datasets
- Accountability

Privacy Challenges

- Patient confidentiality
- Secure storage of medical records
- Compliance with healthcare regulations

Deployment Challenges

- High implementation cost
- Integration with hospital information systems
- Staff training requirements

Legal Challenges

- Medical liability
 - Regulatory approval
 - AI governance
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12. Future Scope and Emerging Trends

Future developments include:

- Multimodal AI combining text, images, genomics, and sensor data
- AI-assisted robotic surgery
- AI digital twins for personalized treatment simulation
- Federated learning for privacy-preserving collaboration
- Wearable AI devices for continuous monitoring
- Explainable AI (XAI) to improve clinician trust
- AI-driven drug discovery and precision medicine
- Integration of Generative AI into electronic health record systems

These innovations are expected to make healthcare more predictive, preventive, personalized, and accessible.

13. Conclusion

AI-assisted medical diagnosis represents one of the most impactful applications of Artificial Intelligence in healthcare. By combining Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and Generative AI, these systems can analyze complex medical data, assist clinicians in diagnosis, and improve patient outcomes. While challenges related to privacy, ethics, explainability, and regulation remain, continued research and responsible deployment will enable AI to become an indispensable partner for healthcare professionals rather than a replacement. AI-assisted diagnosis has the potential to improve efficiency, reduce diagnostic errors, and deliver higher-quality healthcare worldwide.

14. References

1. World Health Organization (WHO). *Ethics and Governance of Artificial Intelligence for Health*.
2. Topol, E. (2019). *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*. Basic Books.
3. Nature Medicine. Research articles on AI in medical imaging and clinical decision support.
4. The Lancet Digital Health. AI applications in healthcare.
5. Google Health. AI for medical imaging and diagnostics.
6. Microsoft Azure AI for Healthcare documentation.
7. Mayo Clinic AI research publications.
8. Apollo Hospitals. AI initiatives in healthcare.
9. National Institutes of Health (NIH) – Artificial Intelligence in Medicine.
10. IEEE Xplore Digital Library – Recent papers on AI-assisted medical diagnosis.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.

4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.

8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.
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